Ethan Lam

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Education

Georgia Institute of Technology Atlanta, GA
B.S. Computer Engineering *May 2028*

- Thread: Cybersecurity/Distributed Systems & Software Design
- Relevant Coursework: GPU Graphics Programming, Computer Arch, Adv Malware Reverse Eng, Circuit Analysis
- Awards: Zell Miller Scholarship, Dean's List

Experience

Portfolio Manager | **GTSF Investment Committee - Quant Sector** Atlanta, GA Aug 2025 - Present

- Developed a portfolio controller in Python to coordinate capital allocation, rebalancing, and risk constraints
- Refactored Python backtesting infrastructure to support train/test separation across multiple strategies
- Integrated and deployed two independent trading strategies into a unified Python execution

Projects

Thermal-Vision Post-Processing Shader | Unity, HLSL, C#

- Built a custom depth-based thermal-vision post-processing effect in HLSL, mapping scene depth to a cold-to-hot color gradient with an adjustable max-distance range
- Implemented Sobel edge detection for real-time object outlining, with tunable thickness and depth multiplier
- Added a film-grain noise pass through a custom Unity Post Processing volume/settings class

Software 3D Renderer | Python, NumPy, Matplotlib

- Built a software rasterization pipeline from scratch using NumPy, implementing model/view/projection transforms, perspective projection, backface culling, clipping, depth ordering, and diffuse illumination without a graphics API
- Processed triangle meshes through homogeneous/NDC coordinate transformations and clipping to correctly render geometry intersecting the camera's viewing volume .

Malware Reverse Engineering: PE Worm Analysis | IDA Pro, x86 Assembly

- Reverse-engineered a Windows PE file-infecting worm in IDA Pro, fully annotating x86 assembly to document its infection, encryption, and evasion behavior
- Traced manual PE export-table parsing and runtime API resolution used to conceal imported functions, along with XOR/rotate encryption and polymorphic code generation
- Identified and documented logic bugs in the sample's anti-VM and anti-debug checks

CryptoGame Comparator | Java, JavaFX, REST APIs, JSON (Gson)

- Developed an interactive JavaFX application comparing 1000+ Steam game prices to 1000+ cryptocurrencies
- Integrated CheapShark and CoinLore APIs to fetch live game and cryptocurrency data

Leadership & Activities

President | **Chess Buddies** Atlanta, GA Aug 2022 – May 2024

- Led an organization dedicated to teaching chess to 50+ underprivileged children across Metro Atlanta
- Directed and coordinated a six-person volunteer team to host 60+ events each school year across 2 schools

Awards

2x AIME Qualifier
1st Place, 50-Person Trading Simulation | 3rd Place, Betting Simulation
Top 3 finishes in regional Science Olympiad: Robot Tour, WiFi Lab, and Scrambler
Semifinalist McKinsey Case Competition

Skills

Languages: Java, C, C#, Python, HLSL/ShaderLab, VHDL, x86 Assembly
Tools: Unity, IDA Pro, FPGA, NumPy, JavaFx, REST API, Linux, Git
Interests: Golf, Basketball, Pickleball, Music, Poker, Rock Climbing